

Figure 9: Tracert – outgoing WAN

To ensure the correct DNS resolution after failover, configure publicly accessible DNS servers under *System – General Setup*.

- Configure public DNS servers such as Level 3 communications DNS 4.2.2.1 or Google DNS 8.8.8.8.
- In addition, select the following checkbox: ‘Allow DNS server list to be overridden by DHCP/PPP on WAN’.

Verify DNS configuration under *Status – Interfaces* to confirm that DNS settings have entries corresponding to ISP DNS servers and to the two additional servers configured earlier.

Following these six steps correctly will configure pfSense 2.3 in dual WAN failover mode.

Verifying dual WAN functionality

Simple verification: Remove cables from WAN/OPT1 interfaces alternately and verify that you can browse the Internet. Here, you may check the IP address of your WAN connection by using the *whatismyip.com* website or by keying in ‘What is my IP’ in Google.

Detailed verification: From the *Status – Gateway* menu, identify and note down the ‘Gateway’ corresponding to the WAN and OPT1 interfaces.

Use the *tracert* (Windows) or *tracpath* (Linux) commands towards any external IP address such as 4.2.2.1 or 8.8.8.8, or even your website IP address. The first hop in the output of this command will be the pfSense IP address, and the second hop will be the gateway of the ISP, indicating whether the traffic is going out from the WAN or OPT1.

Also observe that whenever WAN and OPT1 are both online, traffic is going out via the WAN interface – which is defined as the Tier1 interface. Try different scenarios by changing the tiers of internet connectivity interfaces and observe the results.

Important pfSense menus for diagnostics and troubleshooting

- Status – Interfaces* details the current status of all interfaces (LAN, WAN and OPT1) in this configuration

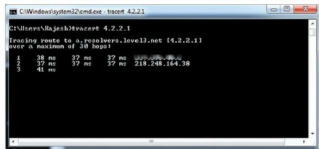


Figure 10: Tracert – outgoing OPT1

- Status – Gateways* provides the current status of both WAN gateways
- Status – Gateways – Gateway Groups* gives the current status of the Gateway Groups. Here, if both the WAN connections are up and running, status of both gateways will show online.
- Status – Services* shows the status of running services. This menu will also enable stopping/restarting of services.

Security best practices

- pfSense uses ‘admin’ as the default user name. As a security best practice, configure an additional user name from the *System – User manager* menu with admin privileges. Further, disable the admin user name by selecting ‘This user cannot login’.
- pfSense has four hours default time to expire idle management sessions. From the *System – User manager – Settings* menu, consider reducing this to five to ten minutes.
- It is also possible to use Microsoft Windows AD authentication for pfSense login from the *System – User manager – Authentication Servers* menu. An excellent article for this is available at <https://forum.pfsense.org/index.php?topic=44689.0>. Keep a watch on this column for more information on the pfSense firewall. 

References

- https://doc.pfsense.org/index.php?title=2.3.1_New_Features_and_Changes
- <https://doc.pfsense.org/index.php/Multi-WAN>
- <https://forum.pfsense.org/index.php?topic=38882.0>
- <https://forum.pfsense.org/index.php?topic=37083.0>
- <https://forum.pfsense.org/index.php?topic=74618.0>
- http://www.squid-cache.org/Doc/config/tcp_outgoing_address/

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